Batch Gradient Descent

Consider

CGPA	IQ	LPA
8.1	93	3.2
7.5	95	3.5

(2, 3)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

(LPA) (CGPA) (iq)

step 1 - Random values of $\beta_0, \beta_{\ldots}$

$\beta_0 = 0, \quad \beta_1, \beta_2 = 1$

Step 2 - epochs = 100, Lr = 0.1

$\beta_0 = \beta_0 - \eta \text{ slope}$

$\beta_1 = \beta_1 - \eta \text{ slope}$

$\beta_2 = \beta_2 - \eta \text{ slope}$

slope is calculated by taking partial derivatives of loss function

i.e. $\frac{\partial L}{\partial \beta_0}$, $\frac{\partial L}{\partial \beta_1}$, $\frac{\partial L}{\partial \beta_2}$

& in case of n-dimensional data We need to calculate (n+1) derivatives

Derivative of Loss Function w.r.t. intercept term

$$L = \frac{1}{n} \sum_{i=1}^{n} (y - \hat{y})^2 \Rightarrow MSE$$

In our dataset we have 2 rows & 2-col 1-target

$$\therefore \quad L = \frac{1}{2}\left[(y_1-\hat{y}_1)^2 + (y_2-\hat{y}_2)^2\right]$$

$$= \frac{1}{2}\left[(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})^2\right]$$

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X Y

x_{11} x_{12} y_1

x_{21} x_{22} y_2

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12}$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22}$$

Partially differentiating w.r.t. β_0

$$\frac{\partial L}{\partial \beta} =$$

$$\frac{1}{2}\left[2(y_1-\hat{y}_1)(-1) + 2(y_1-\hat{y}_2)(-1)\right]$$

$$= \frac{-2}{2}\left[(y_1-\hat{y}_1)+(y_2-\hat{y}_2)\right]$$

$$\therefore \frac{-2}{n}\left[(y_1-\hat{y}_1)+(y_2-\hat{y}_2)+\dots+(y_n-\hat{y}_n)\right]$$

$$\boxed{L = \frac{-2}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i) = \frac{\partial L}{\partial \beta_0}}$$

$$L = \frac{1}{n}\left[(y_1-\beta_0-\beta_1 x_{11}-\beta_2 x_{12})^2 + (y_2-\beta_0-\beta_1 x_{21}-\beta_2 x_{22})^2\right]$$

Now,

Differentiating w.r.t. β_1

$$\frac{\partial L}{\partial \beta_1} = \frac{1}{n}\left[2(y_1-\beta_0-\beta_1 x_{11}-\beta_2 x_{12})(-x_{11}) + 2(y_2-\beta_0-\beta_1 x_{21}-\beta_2 x_{22})(-x_{21})\right]$$

$$= \frac{-1}{n}\Big[\, 2x_{11}(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12}) + 2x_{21}(y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})\Big]$$

$$= \frac{-2}{n}\Big[(y_1 - \hat{y}_1)(x_{11}) + (y_2 - \hat{y}_2)(x_{21})\Big]$$

$$= \frac{-2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)\,x_{i1}$$

here x_{i1} is

β_1	β_2	
x_1	x_2	y
8.1	93	3.2
7.5	95	3.5

$\Uparrow$

$$\begin{bmatrix} x_{11} \\ x_{21} \\ x_{31} \\ \vdots \end{bmatrix}$$

So we are calculating loss function for β_1 $\therefore$ x_{i1}

$\times n_1$

$$\therefore \boxed{\frac{\partial L}{\partial \beta_2} = \frac{-2}{n} \sum (y_i - \hat{y}_i) x_{i2}}$$

Hence, for m columns

$$\boxed{\frac{\partial L}{\partial \beta_m} = \frac{-2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i) x_{im}}$$

When we calculate loss w.r.t any coefficient so we have to multiply x_{im} (particular coeff's corresponding column's value)